FINAL PROJECT

Course: MERN Stack Web Development

Assessment Type: Final Project (Full Stack Application)

Total Marks: 100

Duration: 2–3 Weeks

📌 Project Title

Full Fledged MERN Stack Learning Management System

📖 Project Objective

Students are required to develop a **complete Full Stack MERN Application** that demonstrates:

- Frontend development using React JS
- Backend development using Node.js & Express
- Database integration using MongoDB & Mongoose
- RESTful API creation
- Authentication & Role-Based Authorization

- Proper project architecture

This project must reflect **real-world industry standards.**

🛠 Required Technologies

Frontend

- React JS
- React Router
- Axios
- CSS / Bootstrap / React Bootstrap

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication
- Bcrypt
- Dotenv

👥 User Roles (Mandatory)

The system must support **three roles**:

1️⃣ Admin

- View all users
- Delete users
- Manage courses
- View analytics

2️⃣ Instructor

- Create course
- Edit course
- Delete course
- Upload lessons

3️⃣ Student

- Register & Login
- Browse courses
- Enroll in courses

- View enrolled courses

📄 Required Pages (Frontend)

Public Pages

- Home Page
- About Page
- Course Listing Page
- Course Detail Page
- Login Page
- Register Page

Dashboard Pages (Role-Based)

Student Dashboard

- My Courses
- Profile Page

Instructor Dashboard

- Create Course

- Manage Courses
- Upload Lessons

Admin Dashboard

- Manage Users
- Manage Courses
- Reports / Analytics

🔌 Backend API Requirements

Students must implement RESTful APIs:

Authentication

- POST /register
- POST /login

Course Management

- GET /courses
- POST /courses
- PUT /courses/:id

- DELETE /courses/:id

User Management

- GET /users
- DELETE /users/:id

Enrollment

- POST /enroll
- GET /my-courses

🗄 Database Requirements

Students must create:

User Model

- name
- email
- password
- role
- timestamps

Course Model

- title
- description
- instructor reference
- category
- price

Enrollment Model

- student reference
- course reference
- progress

🔐 Authentication Requirements

- Password hashing using Bcrypt
- JWT-based authentication
- Protected routes
- Role-based middleware

🗁 Project Structure (Mandatory)

Students must maintain proper folder structure:

Backend

- models
- controllers
- routes
- middleware
- config
- server.js
- .env

Frontend

- components
- pages
- services
- context (optional)
- routes

📊 Marking Scheme (100 Marks)

Criteria	Marks
UI/UX Design	15
React Implementation	15
Backend API Development	20
Database Design	15
Authentication & Security	15
Role-Based Functionality	10
Code Quality & Structure	5
Deployment & Testing	5
Total	**100**

📦 Submission Requirements

Students must submit:

1. GitHub Repository Link
2. Live Deployment Link (if deployed)
3. README file with:
    - Project overview

    - Installation steps
    - Technologies used
4. Screenshots of major modules

❌ Important Instructions

- Plagiarism will result in zero marks.
- Code must be properly structured.
- Hard-coded credentials are not allowed.
- Environment variables must be used.
- Proper error handling is mandatory.

🎯 Learning Outcomes

By completing this assessment, students will demonstrate:

- Complete MERN stack development
- Full-stack integration
- Authentication & authorization
- Real-world project workflow

- Industry-level coding practices

Student Declaration

I confirm that this project is my own work and I have not copied it from any unauthorized source.

Student Name: _____
Signature: _____
Date: _____